

Autonomous Robotic-Arm Docking System for Spacecraft

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Abstract—As space becomes more industrialized, Spacecraft docking remains one of the most precision-demanding operations in human spaceflight, traditionally relying on autonomous guidance, navigation, and control (GNC) systems that consume significant propellant and offer limited abort authority. This paper presents the design and development of an autonomous robotic-arm docking system intended to complement, not replace, the NASA International Docking System Standard (IDSS). Building on the heritage of the Androgynous Peripheral Attach System (APAS) and the Low Impact Docking System (LIDS), the proposed system introduces a Revolute-Prismatic-Revolute (RPR) manipulator arm mounted to the docking port of the host spacecraft, capable of capturing and aligning an approaching vehicle with enhanced control authority and reduced propellant dependence. The system integrates the novel application of safety-constrained reinforcement learning trained in NVIDIA Isaac Sim to achieve robust, adaptive docking behavior transferable from simulation to physical hardware. A prototype implementation using NEMA 17 stepper motors, LX-16A bus servos, a linear actuator, and a Teensy 4.0 serves as the experimental testbed. Results from simulation training demonstrate convergent reward accumulation, validating the reinforcement learning approach prior to hardware deployment. This work addresses critical gaps in current docking infrastructure, including safe abort capability, compatibility across emerging commercial platforms, and reduced reliance on spacecraft GNC during terminal approach.

Index Terms—autonomous docking, spacecraft rendezvous, robotic arm, reinforcement learning, IDSS, RPR manipulator.

I. INTRODUCTION

SPACECRAFT docking is one of the most operationally demanding and safety-critical maneuvers in human spaceflight. As missions to the ISS become more frequent, and at the dawn of a new age of commercial space stations approaches, the ability for consistent, safe, and hardware agnostic docking systems is critical. Currently, the ISS is the only space station used by the United States and therefore is the working standard. As the industry develops new stations, there is an opportunity to revisit existing solutions and investigate whether adding new additional methods for docking could be beneficial.

II. BACKGROUND AND EVOLUTION OF DOCKING SYSTEMS

A. NASA International Docking System Standard (IDSS)

The current state-of-the-art is the NASA International Docking System Standard (IDSS). Created 2010, the IDSS became NASA's go-to docking interface, with the plan to have it be standardized across all cooperating agencies as the ISS was the only relevant destination for spacecrafts.

The IDSS allows both docking and berthing, and supports both autonomous and piloted docking. It can be androgynous, passive, or active, allowing for docking with all vehicles that design to interface with it. It can adapt to slight misalignments of docking spacecrafts, and can lower the impact of collision during the docking procedure. Overall, the IDSS is a working docking standard that has served the space industry for the past sixteen years. [1]

B. Heritage Hardware

The formation of the IDSS in 2010 was more accurately the combination and reworking of two pieces of Heritage Hardware:

- Androgynous Peripheral Attach System (APAS)
- Low Impact Docking System (LIDS)

1) Androgynous Peripheral Attach System (APAS):

The first of the two predecessors to the IDSS was the APAS. Created in 1975 through U.S. and Soviet cooperation during the Apollo-Soyuz era, APAS became a major docking interface for both NASA and the Soviet Space Program.

Like the IDSS, the APAS used an androgynous peripheral design, replacing the older probe-and-cone system with alignment petals around the docking ring. This allowed either spacecraft to take the active docking role, making it more flexible than earlier single-sided docking systems. It was a robust design that lead to the IDSS's ability to adapt to the approach of different spacecrafts. [2]

2) Low Impact Docking System (LIDS):

The more recent piece of integrated heritage hardware was the NASA Low Impact Docking System (LIDS), which was adapted in 2001 from the 1996 Advanced Docking Berthing System (ADBS). LIDS was developed to simplify docking operations and reduce the risks associated with mating spacecraft.

The impact shock absorption of LIDS was the main feature carried over to the IDSS. Eventually, it evolved into iLIDS and the NASA Docking System, making it the main bridge between older APAS-style docking and the modern IDSS standard.

C. A Case for Exploring Additional Hardware:

In the space and aerospace industries, Heritage Hardware is defined as

“A product (e.g., complex part, unit, assembly, sub-system or system) whose design has previously undergone qualification and flown.” [3]

In other words, heritage hardware is hardware that is standardized due to prior use and proven functionality. It is a convenient and cost-effective way to design a system, and ideal for widespread use cases since the risk is minimal. However, as is true with all things, a past solution may not always be the only or best solution.

There exist pros and cons for using heritage hardware, as shown below:

TABLE I
HERITAGE HARDWARE BENEFITS AND LIMITATIONS.

Industry Benefits	Potential Limitations
Proven reliability from prior flight systems.	Legacy interfaces can constrain new architectures.
Reduced development risk and unknowns.	May require redesign, re-verification, or expensive parts.
Lower cost and shorter development schedule.	Early reuse decisions may limit optimal system design.

Due to the development of commercial space stations such as VAST, Axiom, Orbital Reef, etc, it is an ideal time to revisit the problem of docking and explore additional solutions.

In this paper, we propose testing an additional solution to autonomous docking, not replacing the current infrastructure.

Phasing out heritage hardware while in use is risky, unrealistic, and disruptive. It is very important to note that we are not suggesting the current system does not work or is inadequate, rather, we wish to explore the idea of implementing an additional system to improve it.

III. PROBLEM STATEMENT AND RESEARCH QUESTIONS

When thinking of a solution, we prioritized the following questions:

- How do we ensure safe and reliable docking between spacecraft and stations?
- How do we guarantee a safe abortion if docking conditions become unsafe?
- How do we maintain compatibility as new stations and docking standards emerge?

To answer these, we decided to have a simple 3 DOF robotic manipulator that would extend from the exterior of the spacecraft and grab the space station rather than the current reverse scenario. The torque applied to the spacecraft must be sufficient to move and realign it in a meaningful way, while also being able to grasp test mounts on the IDSS of the ISS.

Current docking methods rely on Guidance, Navigation, and Control (GNC) maneuvers to align with the IDSS. Our

proposal would be that if an arm or arms can grab the IDSS instead, then the spacecraft would be able to:

- Rely on human operation or automation rather than GNC.
- Maintain a rigid connection to the IDSS for safer docking and abortion procedures.
- Dock faster than floating into the IDSS.

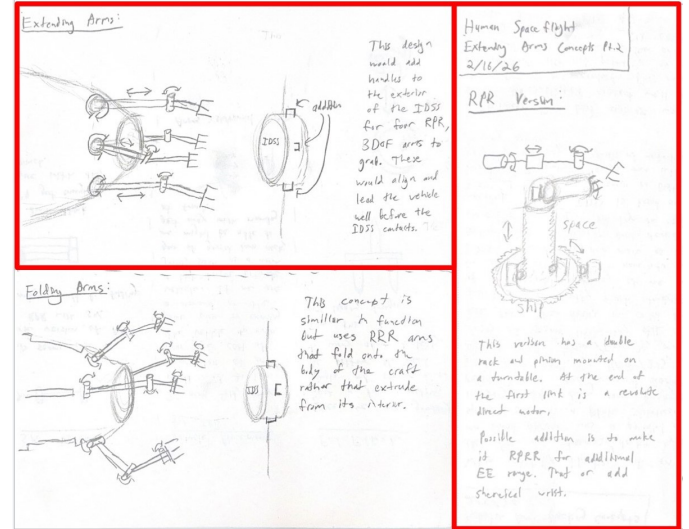


Fig. 1. Initial Sketches of Concept Designs

As seen in figure 1, we considered an Revolute-Revolute-Revolute (R3) arm, and a Revolute-Prismatic-Revolute (RPR) arm. Eventually, we determined that an R3 would need to be too long and would not be able to support the torque needed to function. Therefore, we proceeded to design an RPR arm for our solution.

IV. PROPOSED SYSTEM DESIGN

A. System Overview

Our design proposes attaching latches to an IDSS. The other portion of the design is a Retractable arm attached to the incoming spacecraft. The arm uses a Revolute-Prismatic-Revolute (RPR) configuration allowing appropriate mobility for the arm to be able to grab the latches on the IDSS and maneuver the spacecraft accordingly.

B. Mechanical Design: RPR Manipulator Architecture

The RPR configuration avoids redundant degrees of freedom while ensuring that the latches can be grabbed with precision and basic controls. The rotation at the bottom allows the arm to turn in its Z axis to account for the latch's orientation, while the prismatic joint helps reduce the torque the arm requires by creating a larger moment arm. Finally, the revolute joint allows correction for any remaining misalignment.

The final design of the RPR manipulator is outlined in the standard DH Table and kinematic chain figure below:

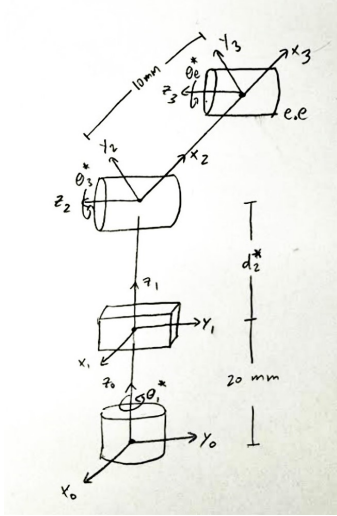


Fig. 2. Denavit-Hartenberg coordinate frame assignment for the RPR manipulator, with z_0 aligned to the stepper rotation axis, d_2^* denoting the prismatic joint displacement, and θ_1^* , θ_3^* representing the revolute joint variables at the base and wrist respectively. Frame origins are placed at each joint following the standard DH convention, with the end-effector frame located 10mm from the wrist joint along x_3 .

TABLE II
DENAVIT-HARTENBERG PARAMETERS

Joints	a	α	d	θ
1 Stepper	0	0	20 mm	θ_1^*
2 Linear actuator	0	0	d_2^*	0
3 Servo	10 mm	90°	0	θ_3^*

C. End-Effector Design

The end effector considers the use of an LX-16A Bus motor. For this specific use, the main challenge is that the servo must be loaded from both ends, with one end being an idler; so far, we have used a geared system to allow for the idler side to be simultaneously driven, which also allows us to incorporate a double gripper, allowing for a stronger grip over the latches.

V. SOFTWARE ARCHITECTURE

The software pipeline for the autonomous docking system spans four principal stages: geometric modeling, physics simulation, reinforcement learning, and real-world deployment. Each stage is tightly coupled to the next, with the overall architecture designed to exploit the unique simplicity of the orbital environment and minimize the manual engineering effort traditionally associated with robotic manipulation in space. Note that here we refer to the spacecraft with the arms as spacecraft A and the spacecraft to be docked to as spacecraft B.

A. CAD Import and Scene Construction

The starting point for the software pipeline is the mechanical CAD model of one of the arms. The simulation captures the

relative positioning of one arm in the docking system with respect to the central camera and the intended target handle on spacecraft B. This model is imported into NVIDIA Isaac Sim, a physically-based robotics simulation platform built on the Universal Scene Description (USD) format. During import, rigid body properties, collision meshes, and joint definitions are derived directly from the CAD assembly, ensuring that the simulated kinematics faithfully reflect the as-designed hardware. The arm is represented as a kinematic chain of three joints, two revolute joints with axes oriented across the spacecraft body and a prismatic joint providing linear extension along the arm, keeping it consistent with the mechanical design. The camera is registered as a sensor in the scene and positioned at the intended centre of the docking face between the arm bases.

B. Simulation Environment

Isaac Sim provides the physics engine, rendering pipeline, and sensor simulation required to reproduce the docking scenario. The simulated environment models contact dynamics at the gripper-handle interface, joint limits, controllers and camera imagery with configurable noise and lens parameters. Critically, the space environment brings significant advantages over terrestrial robotic applications. The near-empty environment means there are no obstacles, occlusions, or unpredictable background elements to contend with. Aside from the regular changes in the lighting due to being in orbit, everything else (in particular, all of spacecraft B being viewed by the camera) is completely standardized. The visual scene is therefore highly constrained, with the camera observes only the arms and the target spacecraft against an extremely predictable background, dramatically reducing the perceptual complexity the policy must handle and narrowing the overall sim-to-real gap. Randomization of lighting direction (the main source of visual change during orbit), surface reflectance, and minor geometric perturbations are applied during training to improve robustness, but the degree of domain randomization required is modest relative to cluttered terrestrial environments. The other key advantage here is the flexibility of the training setup. If any of the hardware or the environment is changed in a future application, it is far easier to modify and retrain in Isaac Sim than adapting other methods such as inverse kinematics with a depth camera.

C. Perception and State Estimation

The onboard camera, centered between the three arm bases on the docking face of Spacecraft A, provides the primary sensing modality. Its field of view is sized to encompass all three arms in their range of motion as well as the three handles on Spacecraft B. Our policy learns by only taking the simulated camera view as input and then comparing the output end-effector positions to the ideal position where they would be grasping Spacecraft B. We can use this comparison (effectively distance between end-effectors and handles, among other signals) as a reward function to minimize via training. No dedicated state-estimation module is needed since the policy is trained to work off just camera input and the ground truth can



Fig. 3. A single manipulator with an example target (grey cylinder) in the Isaac Sim environment

be calculated via simply measuring the distance between end-effectors and handles in the simulated environment. The handle design is rigid, geometrically distinct protrusions attached directly to the hull of Spacecraft B, chosen in part to make this visual tracking problem well-conditioned.

D. Reinforcement Learning

Policy training is performed within Isaac Sim using reinforcement learning. The agent controls the joint commands of all three R-P-R arms simultaneously, although this initial proof of concept is limited to a single arm. The objective is to bring each gripper into contact with its corresponding handle and achieving a secure mechanical capture. The reward signal combines a proximity term that measures the grippers distance from their target handles, a capture term triggered by successfully holding all three handles, and a smoothness penalty on joint accelerations to discourage aggressive motions that could induce structural loads or spacecraft tumbling. The simplicity of the environment and the task makes the reward relatively robust: there are no contact-rich intermediate states or narrow passages to add complexity or required coordination. Training is conducted across a large number of parallel simulation instances to achieve sample efficiency, with initial approach conditions randomised over the possible variation of relative positions, velocities, and orientations. The number of possible variations are relatively small given that this system is simply designed as a complement to GNC systems which would get the spacecraft mostly in the right ballpark for the final approach already.

E. Sim-to-Real Transfer

Once a policy has converged in simulation, deployment to the physical prototype proceeds via direct policy transfer. The fidelity of the Isaac Sim physics engine, combined with the benign sim-to-real conditions inherent to the orbital environment, is expected to yield transferable behaviour without extensive additional fine-tuning. Joint-level controllers on the physical arms receive commands from the same policy network that was trained in simulation, with the camera feed from the real hardware substituted directly for the simulated

image observations. Any residual gap between simulated and real contact dynamics at the gripper-handle interface will be characterized during hardware testing and addressed through targeted domain randomization or a small number of real-world adaptation steps if required. The key adaptation steps are translating the joint angles from the simulator to the real motors (the position of zero radians and positive rotation direction may be different) and emulating the space environment on the prototype, specifically in the lighting conditions.

VI. MODEL AND EXPERIMENTAL PREPARATION

This section details the mechanical design, physical prototype, and electrical implementation of the single-arm proof-of-concept. The prototype was developed iteratively, with CAD modeling in Onshape preceding component procurement and assembly. All hardware was sourced from existing inventory given budget constraints, and the electrical system was validated by independently testing each actuator before integrating them into a unified control loop.

A. CAD Model

The CAD Model was created in Onshape and consisted of a single major assembly consisting of four subassemblies, one for each actuator. The bottom consisted of a platform on which the bottom stepper motor could sit and a flange to hold the stepper in place. The stepper motor allows rotation about the axis normal to the end effector. We designed the next subassembly to be a container to hold the linear actuator. We designed the bottom of the container to mount to a hub attached to the stepper motor. The container holds the linear actuator straight, allowing it to extend in a linear motion and to reduce the torque required by the end effector. A custom adapter was designed to connect the linear actuator to the next actuator, which was the first bus servo. The bus servo subassembly consisted of the linear actuator to bus servo mount, the bus servo, and the shafts connected to the servo's actuator. This servo served as a wrist to fix slight misalignments of the arm when approaching the target. The end effector was designed to be gear based and driven by another servo, where we have a 4-gear system, each driving a claw, creating a 4-claw system for a greater grabbing area and better hold over the latches.

B. Physical Prototype

The Physical prototype was printed in multiple colors, where the base was made first, and heat-set inserts were used to help with screw connections. The flange was screwed on top of the Nema 17 stepper motor, allowing the motor to securely move the other components without tipping or shifting. A hub was secured to the stepper motor's shaft using a set screw, where the linear actuator was connected in the container. The printing of this was difficult as we had to work with printed tolerances to ensure the linear actuator was a tight fit to avoid a slight tilt or angle. There were a few iterations that snapper but finally after using consistent tolerancing, a secure portion was created. We then printed an adapter that slightly forms

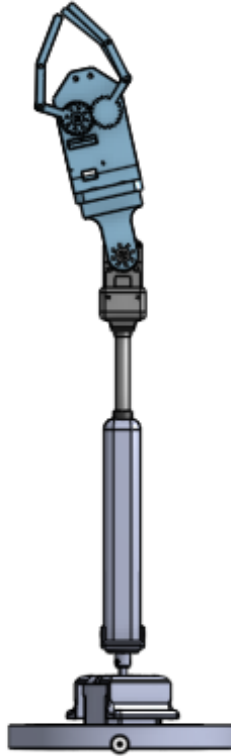


Fig. 4. The full CAD model of the arm.

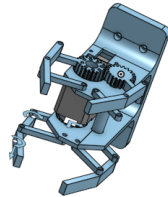


Fig. 5. The end effector consisting of a servo and 4 grippers.

around the shaft of the actuator to avoid tipping of the next component (the LX-16A Bus Servo). The servo controlled the wrist, which printed with relatively low error. Due to this, no additional iteration on this component was necessary. The end effector was the most complicated component due to it needing a lot of parts. We had a platform to mount the servo and each gear. The servo would move one gear, which would move another gear, which in turn was rigidly connected to a gear on the bottom. The bottom gear drives the last gear; it is a four-gear system with each gear driving a gripper. The iterations included making sure the gears were compatible, along with needing to increase the thickness of the grippers for better grip. The system overall due to the careful design, worked well together.



Fig. 6. Full printed prototype of the arm

C. Electrical System and Circuit Implementation

The electrical System is centered around a Teensy 4.0 micro controller [4] powered by a 12V Lithium battery, which feeds all subsystems through a shared breadboard power distribution rail. A common ground across all components is maintained throughout, critical for reliable logic-level communication between boards operating at different voltage levels.

The Two LX-16A bus servos are interfaced through a BusLinker V2.5 [5] connected to the Teensy's hardware serial port at 115200 baud. Since the LX-16A operates between 6-8.4 V [6], a buck converter steps the 12V down before feeding the BusLinker. The servos are daisy-chained on the bus and assigned unique IDs for independent position control. The NEMA-17 stepper [7] and linear actuator [8] are each driven by dedicated L298N H-bridge modules [9], both drawing 12V directly from the battery rail with control signals sourced from Teensy's digital output pins. ENA and ENB are jumpered high on both driver boards. The complete wiring topology is illustrated in Fig. 7. The dominant current draws are the stepper motor at up to 2A per phase [7] and the LX-16A servos at 3A stall current [6], both of which are within the supply headroom of the 12V battery under normal operating conditions.

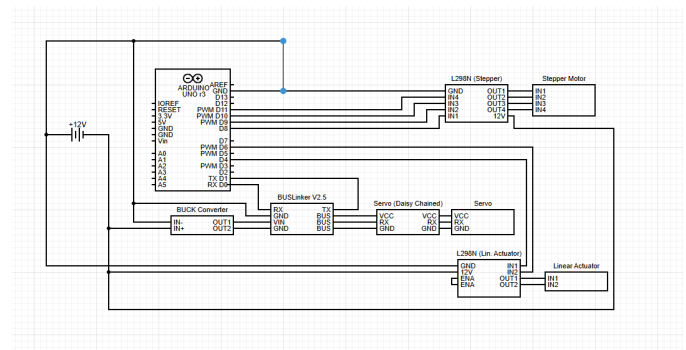


Fig. 7. Circuit diagram illustrating the electrical connections

VII. POTENTIAL LIMITATIONS

Several practical constraints shaped the scope and fidelity of this prototype. From a budget perspective, the project operated with zero dedicated funding; all hardware was sourced from spare parts and preciously owned components. While this demonstrates the high feasibility of the concept, it also signifies that component selection was driven by availability rather than optimal specifications. The linear actuator, for instance, exhibited mechanical dilation under off-axis loading, a limitation that would not be acceptable in a flight-ready system where actuator reliability is paramount.

Furthermore, environmental factors present perhaps the most significant gap between the prototype and a flight system. The prototype was tested under standard laboratory conditions with no attempt to replicate the thermal, vibrational, or microgravity environment of orbit. Lighting conditions, which are primary source of visual variability in the sim-to-real pipeline, were not controlled during hardware testing. Additionally, the mechanical tolerances achievable with 3D-printed components are substantially lower than what would be required for structural reliability in a launch and orbital environment.

With that being said, the current system has no sensor feedback. No force or torque sensor at the gripper interface, no practical camera for RL simulation implies inadequate information for advanced control logic for the docking arms. All things that would've been considered given more generous timeline as opposed to a semester-long project.

VIII. FUTURE WORK AND NECESSITIES

The most immediate extension is expanding the single-arm prototype to the full three arm configuration, which is largely a replication effort given the existing architecture already supports scaling. More meaningfully, several hardware upgrades would significantly improve system capability. The linear actuator should be replaced with a ball-screw driven prismatic joint to handle off-axis loading, and upgrading torque-sensing servos would enable compliant contact control at the gripper interface. The end effector would also benefit from fingertip force sensors to confirm secure capture rather than purely relying on position.

On the software and deployment side, migrating from the Teensy 4.0 to a Raspberry Pi and integrating an onboard camera would enable the full sim-to-real pipeline to be exercised on hardware. Ultimately, a credible evaluation of this system would require a dedicated test environment with controlled lighting, a suspended target spacecraft mock-up to simulate microgravity dynamics, and a motion platform to introduce realistic approach perturbations: the essential bridge between a prototype and a viable system for spaceflight consideration.

IX. CONCLUSION

This paper presents the design and early-stage development of an autonomous robotic-arm docking system intended to complement existing spacecraft docking infrastructure. The RPR manipulator architecture introduces mechanical control

authority during terminal approach, the phase most vulnerable to propellant exhaustion, sensor noise, and limited abort options under conventional GNC-dependent systems. As low Earth orbit becomes increasingly commercialized and crewed missions proliferate across providers with varying GNC maturity, the docking interface represents a critical junction where failure carry irreversible human consequences. The proposed system addresses this not by replacing autonomous GNC, but by providing a physically-grounded redundancy layer through which an operator or autonomous policy can intervene, correct, or abort independent of the approaching vehicle's onboard systems.

Simulation results demonstrate convergent reward accumulation under the reinforcement training regime, validating control approach prior to hardware deployment. The prototype confirms that the RPR architecture is mechanically realizable at low cost and that the various actuators can be coordinated from a single microcontroller. Configuring the full system potential requires expanding to the three-arm configuration, upgrading compute and sensing hardware to enable sim-to-real transfer, and validating performance under representative environmental conditions. Each of these show a tractable next step toward a system that could meaningfully reduce risks at one of human spaceflight's most consequential interfaces.

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